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Assignment 4

① Static RAM (SRAM) uses flip flops to store data, making it faster, more reliable and expensive. It is mainly used in cache memory.

Dynamic RAM (DRAM) uses capacitors to store data, making it slower and cheaper and required periodic refreshing. It is used in main memory.

② Locality of reference is the principle that programs tends to access a small portion of main memory repeatedly. It includes -

- Temporal locality : Recently accessed data is likely to be accessed again

- Spatial locality : Nearby memory locations are likely to be accessed.

③ Cache memory is used to reduce the average memory access time by storing frequently accessed data closer to the CPU.

4) Cache Indexing mechanisms determine how memory blocks are mapped to cache.

- Directly mapped
- Set associative
- Fully associative.

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(5) A large cache block size improves spatial locality but increases miss penalty and may waste memory bandwidth.

(6) Cache associative refers to the number of locations in cache where a memory block can be placed. Higher associativity reduces conflicts misses.

(7) A larger cache reduces miss rate but increases cost and access latency, leading to a trade off between performance and expense.

(8) Write though → Data is written to both cache and main memory simultaneously.

Write back - Data is written only to cache and updated in main memory when the block is replaced.

(9) Average memory access time (AMAT) is average time required to access memory. It is given by:

AMAT = Hit time + (Miss rate x Miss penalty)

10) Least recently used (LRU) is a replacement policy in which the block that has not been used for the longest time is replaced first.

11) Memory interleaving is a technique that allows simultaneous access to multiple memory modules, thereby increasing memory bandwidth.

12) Tracks are concentric circular paths on the disc surface where data is stored. Sectors are subdivisions of tracks.

that store fixed-size data blocks.

(x) Flash memory is a non-volatile storage technology used in SSDs and USB drives. It is faster than magnetic discs and has no moving parts.

(xi) In memory mapped I/O, input/output devices share the same memory and are accessed using standard memory instructions.

(xii) In I/O mapped I/O, devices have a separate address space and are accessed using special I/O instructions.

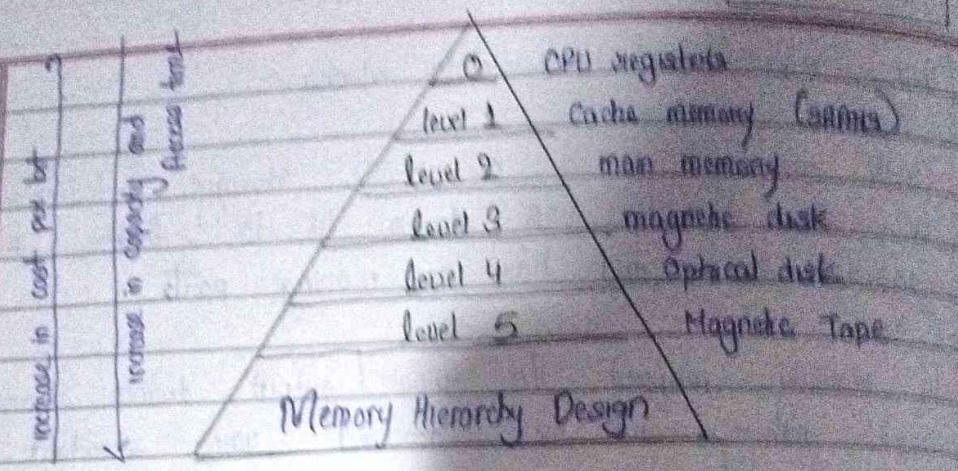
(xiii) Programmed I/O is a method where CPU continuously checks the status of an I/O device and actively controls the data transfer.

(xiv) In interrupt-driven I/O, the device sends an interrupt signal to the CPU when it is ready, allowing the CPU to perform other tasks meanwhile.

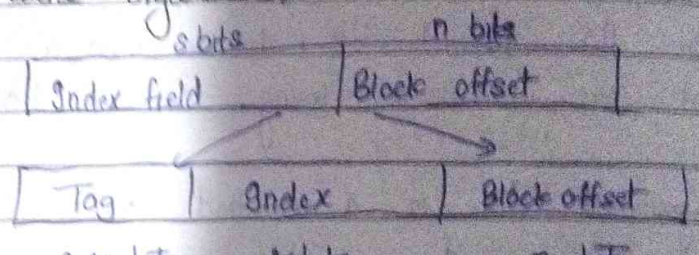
(xv) Direct memory Access (DMA) allows I/O devices to transfer data directly to or from main memory without the continuous CPU involvement, improving system efficiency.

Section B

1) memory hierarchy Design



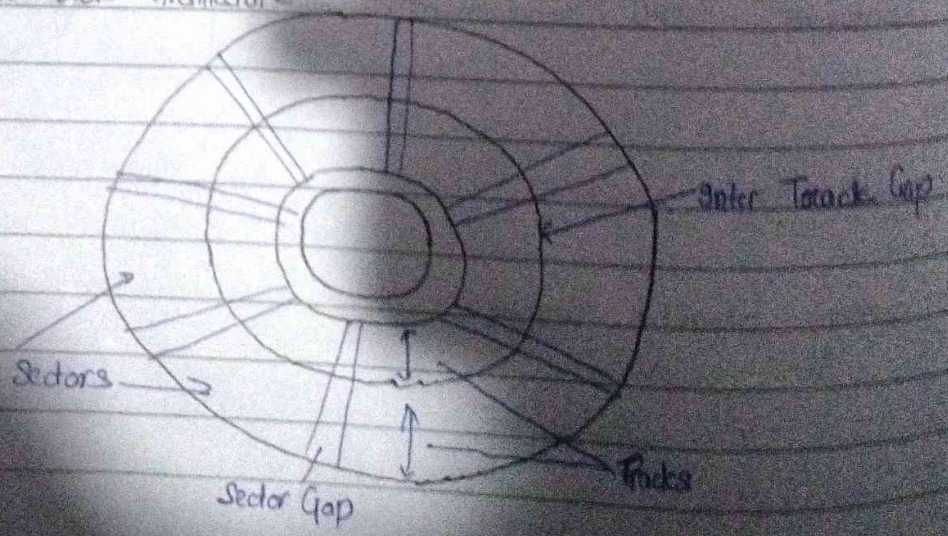
② Direct mapped cache organisation



③ Write back | Write Through

before		after	before		after
x = 100	CPU	x = x + 200	x = 100	CPU	x = x + 200
x = 100	Cache memory	x = 300	x = 1200	Cache memory	x = 300
x = 100	Main memory	x = 300	x = 100	main memory	x = 300

④ Magnetic Disk Architecture



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Section C

1) Temporal locality applies. The program repeatedly accesses the same memory location. The principle says that if a location is referenced, it is likely to be referenced again soon. Caches exploit this by keeping the location in fast memory after first access, thus reducing future access time.

(2) Larger blocks bring more consecutive data, improve spatial locality and reducing compulsory / capacity miss rates. However:

- larger miss penalty: Transfer a bigger block takes more time, increasing the cost of a miss.

- Cache pollution: Useless data inside the block evicts potentially useful data, increasing other misses.

+200 - fewer blocks: For a fixed cache size, larger blocks mean fewer total blocks which may increases conflict misses.

3) During every write operation

- On a write hit - The cache line is updated and simultaneously or immediately after the corresponding main memory location is written. The CPU may stall until the memory line writes completes.

- On a write miss - Common policy is no-write-allocate - the block is not brought into cache; the write goes directly to main memory.

The problem is that main memory is always up to date but bus traffic and latency are high.

(4) Direct memory access (DMA). The memory card includes a DMA engine. The CPU assigns the DMA transfer

source = which card buffer

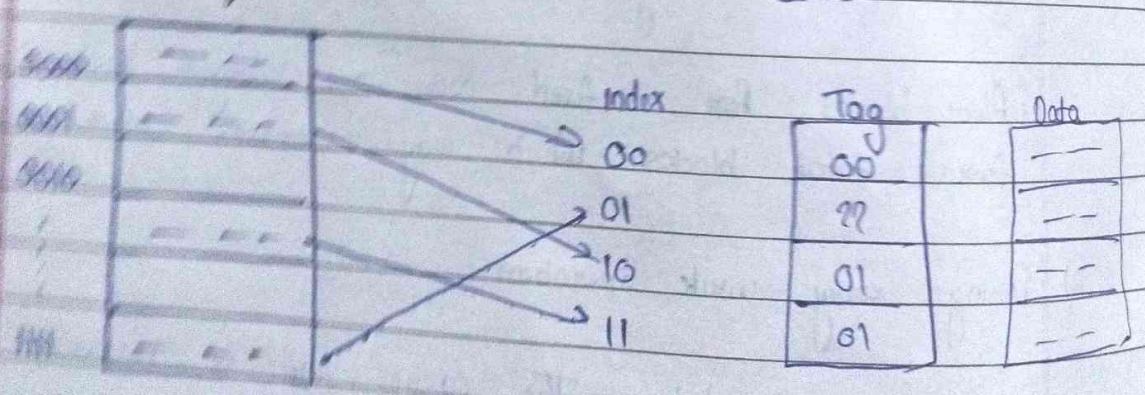
destination = memory address, count and direction

Then the DMA engine moves data directly between the card and memory. The CPU is free to handle other tasks. Upon the completion, the DMA controller interrupts the CPU.

This stops the CPU from heavy data copying and increases system throughput.

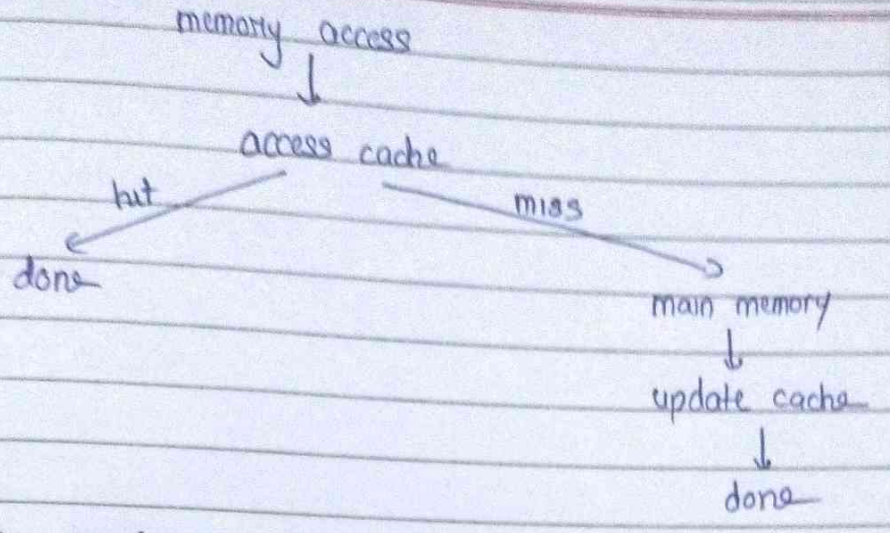
Section D

(1)

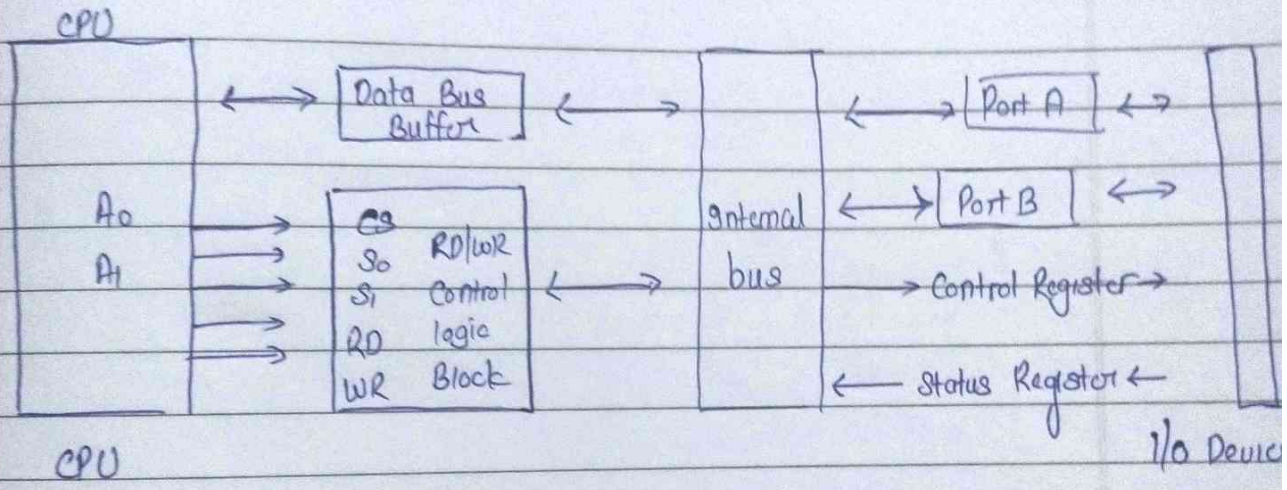


(2) AMAT equation:

AMAT = Hit time + miss rate × miss penalty



③ I/O data Tranfer Technique



Advantage DMA – offloads CPU completely during the bulk data transfer.